Polar Equations and Graphs

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Recognizing polar graphs**

Identify the graph of each polar equation:

a $r = 3$

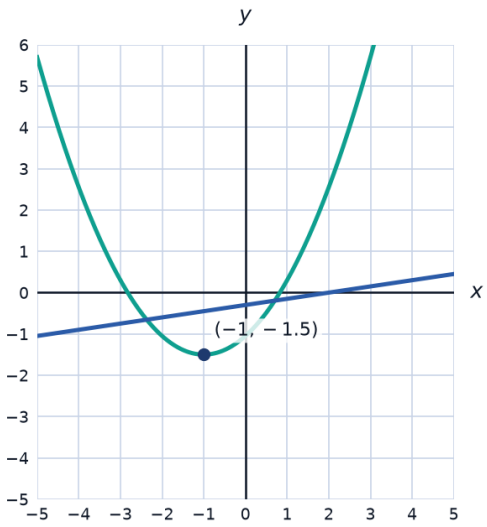
b $\theta = \frac{\pi}{6}$

c $r = 2 + 2\cos\theta$

d $r = 4\sin 3\theta$

2 **[Foundational]** • **Target: Recognizing polar graphs**

For the limaçon $r = 1 + 2\cos\theta$:



a Explain how you know it has an inner loop.

b Find the values of θ in $[0, 2\pi)$ at which the curve passes through the pole.

3 **[Foundational]** • **Target: Recognizing polar graphs**

For the rose $r = 5\sin 2\theta$:

a State the number of petals.

b State the maximum value of $|r|$ and the smallest positive θ at which it occurs.

4 **[Foundational]** • **Target: Recognizing polar graphs**

Determine whether the graph of $r = 2 + \sin\theta$ passes through the pole, and state the type of limaçon.

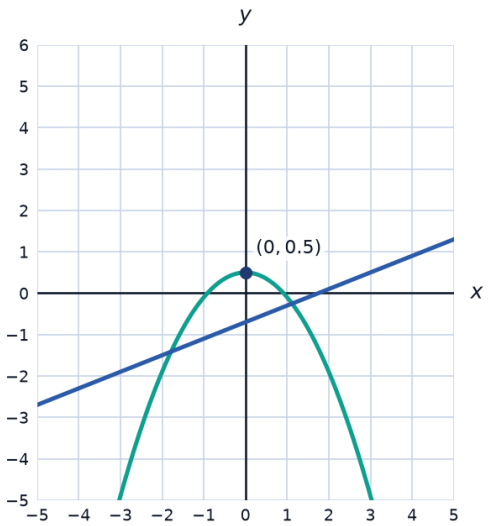
© ClipSAT — clipsat.org | All Rights Reserved

5 [Foundational] • Target: Symmetry tests

Test $r = 3\cos 2\theta$ for symmetry about the polar axis (replace θ with $-\theta$) and about the pole (replace r with $-r$).

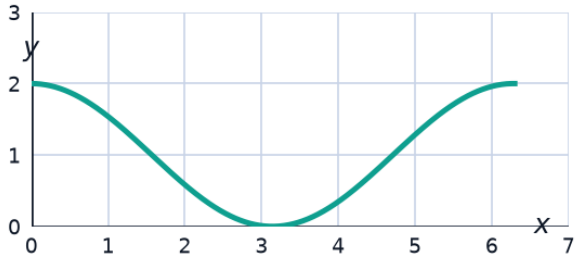
6 [Foundational] • Target: Symmetry tests

Test $r = 2 + 2\sin\theta$ for symmetry about the line $\theta = \frac{\pi}{2}$ (replace θ with $\pi - \theta$).



7 [Foundational] • Target: Graphing from a table of values

For the cardioid $r = 1 + \cos\theta$, complete the table of r-values: $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$.



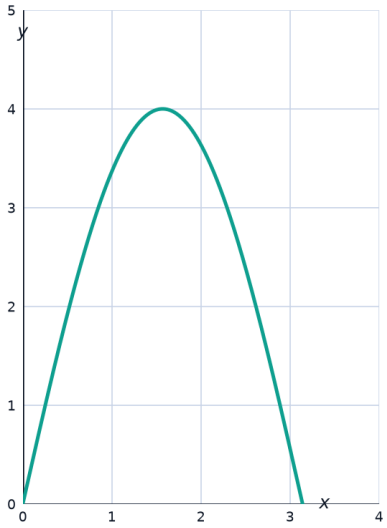
a $r(0)$ **b** $r\left(\frac{\pi}{2}\right)$ **c** $r(\pi)$ **d** $r\left(\frac{3\pi}{2}\right)$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

© ClipSAT — clipsat.org | All Rights Reserved

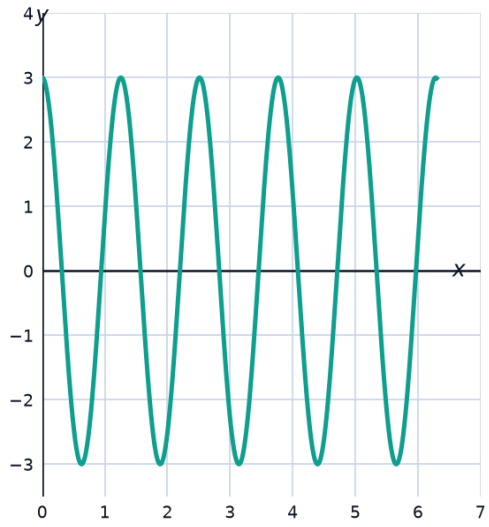
8 **[Intermediate]** • **Target: Graphing from a table of values**

For the circle $r = 4\sin\theta$, find the value(s) of θ in $[0, \pi)$ where r reaches its maximum, and state that maximum value.



9 **[Intermediate]** • **Target: Rose curves in depth**

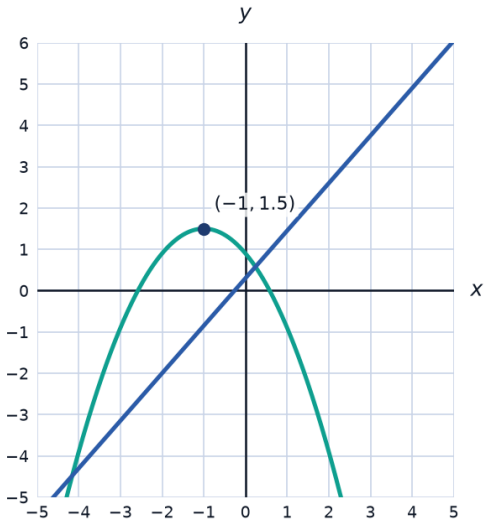
For the rose $r = 3\cos 5\theta$, state the number of petals and the length of each petal (maximum $|r|$).



© ClipSAT — clipsat.org | All Rights Reserved

10 [Intermediate] • Target: Rose curves in depth

Explain, using the general rule for rose curves $r = a\cos(n\theta)$ or $r = a\sin(n\theta)$, why n odd gives n petals but n even gives $2n$ petals.



11 [Intermediate] • Target: Identifying limaçon subtypes

Classify each limaçon $r = a + b\cos\theta$ as having an inner loop, being dimpled, or being convex (roughly oval, no dimple), based on the ratio $\frac{a}{b}$:

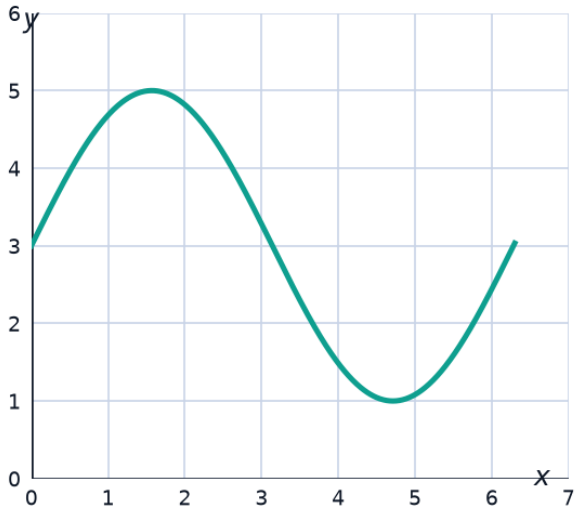
a $r = 1 + 3\cos\theta$

b $r = 3 + 2\cos\theta$

c $r = 4 + \cos\theta$

12 [Intermediate] • Target: Finding maximum and minimum r-values

For the limaçon $r = 3 + 2\sin\theta$, find the maximum and minimum values of r, and the angles at which they occur.



© ClipSAT — clipsat.org | All Rights Reserved

13 [Intermediate] • Target: Comparing sine- and cosine-based polar curves

Explain how the graph of $r = 3 + 3\sin\theta$ compares to $r = 3 + 3\cos\theta$ (same cardioid shape, different orientation).

14 [Intermediate] • Target: Sketching from key features

Without fully graphing, list the key features you would compute first to sketch $r = 2 - 4\cos\theta$ (max/min r, pole crossings, symmetry).

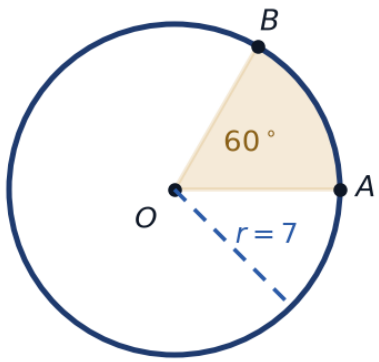
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: A capstone graphing problem • Cross-Lesson: 2.02 Radian Measure and Arc Length

For the rose $r = 4\cos(3\theta)$, find: (a) the number of petals, (b) the maximum value of r, (c) the value(s) of $\theta \in [0, \pi)$ where the first petal reaches its maximum length, and (d) the angles where the curve passes through the pole in $[0, \pi)$.

16 [Challenging] • Target: Distinguishing curve families by equation form alone • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Without graphing, state whether each polar equation is a circle, a line, a rose, or a limaçon, based purely on its algebraic form: $r = 7$, $r = 5\sin(4\theta)$, $\theta = \frac{2\pi}{5}$, $r = 2 - 2\cos\theta$.



17 [Challenging] • Target: Comparing sine- and cosine-based polar curves • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

The curve $y = x^2 - 6x + 5$ can be mapped to $y = x^2$ by a sequence of two geometric transformations. Express the equation in vertex form, state the vertex, and describe fully each transformation.

© ClipSAT — clipsat.org | All Rights Reserved

18 [Challenging] • Target: Sketching from key features • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Convert the polar equation $r = 6\cos\theta - 8\sin\theta$ to Cartesian rectangular form, and identify the geometric curve, its center, and its radius.

19 [Challenging] • Target: A capstone graphing problem • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Analyze the polar curve $r(\theta) = 2 - 4\cos\theta$ for $0 \le \theta < 2\pi$. Identify the curve type, find the maximum distance from the pole, and determine the exact angles θ where the curve passes through the pole.

20 [Challenging] • Target: Distinguishing curve families by equation form alone • Cross-Lesson: 3.03 Sum and Difference Formulas

Find the set of values of the constant k for which the equation $kx^2 + 4x + (k - 3) = 0$ has real roots.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Recognizing polar graphs

The graph of the polynomial function $f(x) = x^3 + ax^2 + bx - 6$ has a local maximum at $x = -3$ and a local minimum at $x = 1$. Find the values of a and b.

a. $a = -3, b = 9$

b. $a = 3, b = -9$

c. $a = -6, b = -3$

d. $a = 6, b = 3$

22 [Foundational] • Target: Symmetry tests

The minimum point of $y = x^2$ is translated by the vector $(-3; 6)$ to give curve C. State the coordinates of the minimum point of C.

a. (3,6)

b. (6,-3)

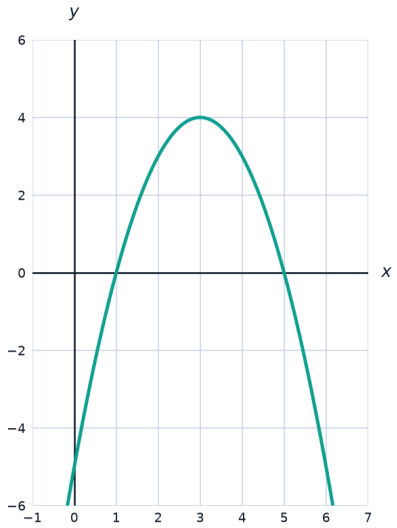
c. (-3,6)

d. (-3,-6)

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Graphing from a table of values

The graph of the function $f(x) = -(x-3)^2 + 4$ is shown in the xy-plane below. What is the maximum value of $f(x)$?



a. -5

b. 3

c. 4

d. 7

24 [Foundational] • Target: Rose curves in depth

The complex number z satisfies the equation $|z-(6+8\mathrm{i})| = 4$. Given that z^* denotes the complex conjugate of z, determine the maximum possible value of $|z-\mathrm{i}z^*|$.

a. $8+2\sqrt{2}$

b. $8-2\sqrt{2}$

c. $4\sqrt{2}+2$

d. $4+\sqrt{2}$

25 [Foundational] • Target: Identifying limaçon subtypes

Consider the rational equation $\frac{x+1}{x-3} + \frac{k}{x+1} = \frac{x+13}{x^2-2x-3}$ where k is a constant. What is the sum of all real values of k for which the equation has no real solution?

a. -10

b. -7

c. -3

d. 10

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Finding maximum and minimum r-values

What are the values of x that satisfy the equation $|3x - 2| = 2|x + 1|$?

a. $x = 0$ and $x = 4$

b. $x = 0$ and $x = -4$

c. $x = \frac{1}{5}$ and $x = 4$

d. $x = \frac{1}{4}$ and $x = \frac{3}{2}$

27 [Foundational] • Target: Comparing sine- and cosine-based polar curves

A factory produces two types of components, A and B, using two machines, M1 and M2. The time (in minutes) required to produce one unit of each component on each machine is given in the table below: | Component | Machine M1 | Machine M2 | |---|---|---| | A | 2 | 3 | | B | 4 | 1 | The factory has 480 minutes of M1 time and 240 minutes of M2 time available per day. Let x be the number of units of component A produced per day and y be the number of units of component B produced per day. Suppose the profit from selling one unit of component A is 5 and the profit from selling one unit of component B is 8. The factory wants to maximize its total daily profit. Formulate this problem as a linear programming problem, defining the objective function and the constraints.

a. Maximize $P = 5x + 8y$ subject to $2x + 4y \le 480$, $3x + y \le 240$, $x \ge 0$, $y \ge 0$.

b. Maximize $P = 8x + 5y$ subject to $2x + 3y \le 480$, $4x + y \le 240$, $x \ge 0$, $y \ge 0$.

c. Maximize $P = 5x + 8y$ subject to $4x + 2y \le 480$, $y + 3x \le 240$, $x \ge 0$, $y \ge 0$.

d. Maximize $P = 5x + 8y$ subject to $2x + 4y \le 240$, $3x + y \le 480$, $x \ge 0$, $y \ge 0$.

28 [Foundational] • Target: Sketching from key features

The graph of the polynomial function $f(x) = x^3 - ax^2 + bx - 6$ has a local maximum at $x = -1$ and a local minimum at $x = 3$. Determine the value of $a + b$.

a. 5

b. -5

c. 1

d. -1

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: A capstone graphing problem

A research team is studying the impact of different fertilizer types on crop yield. They conducted an experiment with two fertilizer types, A and B, and recorded the yield in kilograms per plot. The data is summarized in the two-way table below. The total number of plots is 200. | Fertilizer Type | Yield < 50 kg | Yield $\geq$ 50 kg | Total | |---|---|---|---| | A | 30 | 70 | 100 | | B | 50 | 50 | 100 | | Total | 80 | 120 | 200 | Given that a plot was selected at random and its yield was less than 50 kg, what is the probability that it received fertilizer type B? Provide your answer as a fraction in its simplest form.

a. 1/2

b. 5/8

c. 3/8

d. 1/4

30 [Intermediate] • Target: Distinguishing curve families by equation form alone

The graph of the cubic polynomial $P(x) = x^3 + ax^2 + bx + c$ has a local maximum at $(-1, 8)$ and a local minimum at $(3, -24)$. Find the value of a.

a. -3

b. -1

c. 1

d. 3

31 [Intermediate] • Target: Recognizing polar graphs

Consider the function $g(x) = x^2 - ax + b$. If the graph of $y = g(x)$ has a minimum value of -1 at $x = 3$, find the values of a and b.

a. $a = 6, b = 8$

b. $a = -6, b = 8$

c. $a = 6, b = -1$

d. $a = -6, b = -1$

32 [Intermediate] • Target: Symmetry tests

Consider a cubic polynomial $P(x) = ax^3 + bx^2 + cx + d$. The graph of $P(x)$ has a y-intercept at $(0, 12)$. It is known that $P(x)$ has a local maximum at $x = -1$ and a local minimum at $x = 3$. If the value of the local maximum is 20, determine the value of the constant term d of the polynomial.

a. 12

b. 10

c. 8

d. 6

© ClipSAT — clipsat.org | All Rights Reserved

33 [Intermediate] • Target: Graphing from a table of values

Express $\frac{1+\cos\theta+i\sin\theta}{1+\cos\theta-i\sin\theta}$ in polar form.

a. $e^{i\theta}$

b. 1

c. $e^{2i\theta}$

d. $\cos\theta+i\sin\theta$ only for $\theta=0$

34 [Intermediate] • Target: Rose curves in depth

Find the maximum value of $f(x)=x^a(1-x)^b$ on $[0,1]$ where $a,b>0$.

a. $\left(\frac{a}{a+b}\right)^a\left(\frac{b}{a+b}\right)^b$

b. $\frac{a^a b^b}{(a+b)^{a+b}}$

c. $\frac{1}{4}$ for $a=b=1$

d. Both A and B

35 [Intermediate] • Target: Identifying limaçon subtypes

If $z_1=2\angle 45°$ and $z_2=3\angle 30°$ (polar form), find $|z_1 z_2|$.

a. $6\angle 75°$

b. 5

c. $\sqrt{13}$

d. 6

36 [Intermediate] • Target: Finding maximum and minimum r-values

Identify the Cartesian curve represented by $r=6\cos\theta$.

a. Circle of radius 6 centered at $(0,0)$

b. Circle of radius 3 centered at $(3,0)$

c. Cardioid with parameter $a=3$

d. Ellipse with semi-major axis 6

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Comparing sine- and cosine-based polar curves

Find the principal cube root of $8i$ in polar form.

a. $2(\cos\frac{\pi}{2}+i\sin\frac{\pi}{2})$

b. $8(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6})$

c. $2(\cos 0+i\sin 0)$

d. $2(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6})$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Sketching from key features • Cross-Lesson: 3.01 The Unit Circle and Exact Values

The curve $y=x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(-2;\ -6)$. State the coordinates of the new vertex.

a. (-2,6)

b. (-2,-6)

c. (-6,-2)

d. (2,-6)

39 [Challenging] • Target: A capstone graphing problem • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

The curve $y=x^2$ is translated by the vector $(7;4)$. State the coordinates of the new vertex.

a. (-7,4)

b. (7,4)

c. (4,7)

d. (7,-4)

40 [Challenging] • Target: Distinguishing curve families by equation form alone • Cross-Lesson: 3.03 Sum and Difference Formulas

Identify the polar curve $r=5+3\cos\theta$.

a. a cardioid

b. a circle

c. a dimpled limaçon

d. a rose with 3 petals

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: Recognizing polar graphs • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

The point $(3, -2)$ lies on the curve $y = f(x)$. Find the coordinates of the corresponding point on $y = f(x + 5) - 2$.

a. (-2,-4)

b. (-2,4)

c. (3,-2)

d. (-4,-2)

42 [Challenging] • Target: Symmetry tests • Cross-Lesson: 4.01 Circles and Parabolas

Convert $r = \frac{4}{1 - 2\cos\theta}$ to Cartesian form and identify its type.

a. Parabola: $y^2 = 4(2)x$

b. Hyperbola: $3x^2 - y^2 + 16x + 16 = 0$

c. Circle: $x^2 + y^2 = 4$

d. Ellipse: $3x^2 + y^2 = 16$

43 [Challenging] • Target: Graphing from a table of values • Cross-Lesson: 4.02 Ellipses

For what values of x does x^x reach a minimum?

a. $x = 0$

b. $x = e$

c. $x = 1$

d. $x = 1/e$

44 [Challenging] • Target: Rose curves in depth • Cross-Lesson: 4.03 Hyperbolas

Find the equation of the line through $(-1, -4)$ with gradient -2, in the form $y = mx + c$.

a. y=2x-6

b. y=-2x-5

c. y=-2x-6

d. y=-2x+6

© ClipSAT — clipsat.org | All Rights Reserved

45 [Challenging] • Target: Identifying limaçon subtypes • Cross-Lesson: 5.01 Vectors: Components and Operations

Determine the type and key features of $16x^2 - 9y^2 = 144$.

a. Parabola

b. Ellipse

c. Hyperbola: $x^2/9 - y^2/16 = 1$, $a = 3, b = 4, c = 5$

d. Circle

© ClipSAT — clipsat.org | All Rights Reserved